

HDR VR

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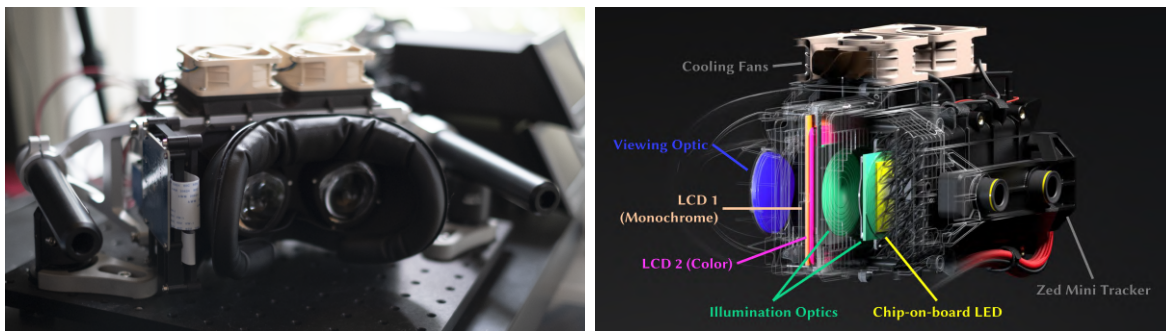


Figure 1: Left: Our prototype high-dynamic-range VR demonstrator. Right: A cutaway view. The LED’s emission cone is shaped by a pair of Fresnel lenses to illuminate color and monochrome LCDs stacked in series, which are imaged by the eyepiece.

ABSTRACT

The human visual system can resolve luminances from near zero to over a million candelas per meter squared (nits), and is able to simultaneously resolve over 4 orders of magnitude without adaptation [Kunkel and Reinhard 2010]. While most traditional displays only replicate a fraction of this range, high-dynamic-range (HDR) displays aim to support ranges closer to perceptual limits [Reinhard et al. 2010] and have achieved widespread use in cinemas and home theaters, but the perceptual impact of HDR remain largely unexplored in the context of Virtual Reality (VR) displays, which are typically limited to peak luminance values below 200 nits [Mehrfard et al. 2019]. To address this, we present an HDR VR demonstrator with a display system comprised entirely of off-the-shelf parts, capable of peak luminances over 16,000 nits. We achieve this without reducing the field of view (FOV) or simultaneous contrast relative to commercially-available VR headsets, while maintaining support for binocular and motion parallax depth cues. Consequently, our prototype has the potential to achieve a higher degree of perceptual realism than existing direct-view devices like HDR televisions and other high-luminance prototypes.

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1 DISPLAY ASSEMBLY

We designed our display with the following targets in mind:

- Exceed 10,000 nits peak luminance
- Match existing VR FOV, resolution, contrast, and framerate
- Support interactive stereoscopic content with 6 degree-of-freedom tracking

We followed existing VR architectures while relaxing power, thermal, and weight limits. See Figure 1 for a photograph and cutaway view of the key headset components, which include

Backlight: We use one Lighten Phoenix LTHX1212-060-005 chip-on-board phosphor LED for each eye, driven by a Thorlabs DC2200 LED power supply. Each LED is followed by an Edmund Optics 13-457 f=10mm Fresnel lens, a 28mm air gap, then an Edmund Optics 43-024 f=38.1mm Fresnel lens to steer the emission cone toward the display eyepiece

Modulation: Following the work of Seetzen et al. [2004], we employ a dual-modulation approach using liquid crystal displays (LCDs). Due to the low efficiency and potential for color moiré patterns when using two color LCDs, we use one color and one

monochrome LCD. Until recently, transmissive LCDs were uncommon, requiring the disassembly of displays with integrated backlights [Rhodes et al. 2019]. The rise in popularity of desktop resin 3D printers has led to the widespread availability of high resolution, high contrast transmissive displays. We use a Wisecoco 6-inch 1620×2560 monochrome display, and a Sharp LS060R1SX02 6-inch 1440×2560 transmissive color display

Viewing Optics: We used viewing optics from a Meta Quest 2 headset to achieve a ~90 degree field of view. The monochrome LCD is placed at the eyepiece’s focal length, with a virtual image conjugate at infinity. The color display is imaged beyond infinity to help reduce moiré effects induced by the two pixel grids.

Thermal Management: Both LCDs have an operating temperature below 60 degrees Celsius, above which the liquid crystals remain in an isotropic state regardless of the electric field and cannot display an image. We employ a pair of 60mm fans to draw cool air from below the headset across the front and back of the LCD stack and a sintered metal heat sink for the LED backlights before exiting the top of the headset.

Tracking: A Zed Mini tracking camera is affixed to the front of the headset to provide tracking.

2 SOFTWARE

Dual-layer HDR decompositions often require a deconvolution step to correct the mismatch in optical resolutions of the two layers, which is particularly important for the local dimming arrays commonly found in HDR televisions. In our case, the circle of confusion for a point on the rear color LCD has a footprint of 1.75 pixels on the front monochrome LCD. Because of this small size and the lower sensitivity of the human visual system to color contrast [Kim et al. 2013], we use a simple factorization without deconvolution. The target color image $[I_r^*, I_g^*, I_b^*]$ is normalized to the floating point intensity range $[0, 1]$. A square-root monochrome image I_m is calculated using Rec709 color coefficients C_r , C_g , and C_b :

$$I_m = \sqrt{C_r I_r^* + C_g I_g^* + C_b I_b^*} \quad (1)$$

Then, the color component I_c is extracted using a naive factorization:

$$I_c = \begin{cases} \left[\frac{I_r^*}{I_m}, \frac{I_g^*}{I_m}, \frac{I_b^*}{I_m} \right], & \text{if } I_m > 0 \\ [0, 0, 0], & \text{otherwise} \end{cases} \quad (2)$$

This process is implemented as a post-processing shader in Unity. The headset is connected via two HDMI cables to an NVIDIA RTX 3090 GPU in a desktop PC. The headset runs in real time at the 50Hz native refresh rate of the two LCDs.

3 RESULTS

We measured the peak brightness and dynamic range of the headset using a Konica Minolta CS-2000A spectroradiometer with a VR lens attachment. The peak brightness, sampled in the center, was measured at 16,632 nits. The dark state was measured at 0.05 nits, for a sequential contrast ratio of 332,640:1. Instantaneous contrast was measured at 39:1 by capturing a checkerboard pattern and its

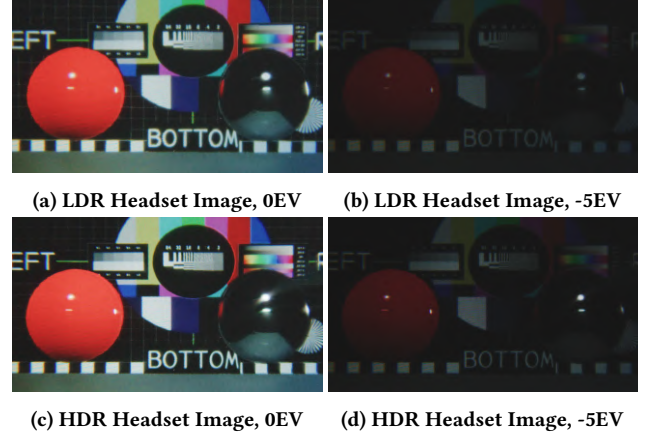


Figure 2: Through-the-lens captures of our prototype headset showing a synthetic scene illuminated with a captured HDR light probe. Top Row: The headset operating in low dynamic range mode. Luminance values are clipped to 100 nits. Bottom Row: An equivalent capture when operating at full dynamic range. The right image in both rows is adjusted down 5 stops to show how specular highlights are preserved in HDR mode.

complement, with checker size slightly larger than the instrument integration area.

Figure 2 depicts the headset in operation using a target HDR virtual scene created in Unity using the real-time decomposition described in Section 2. The top row shows a low dynamic range operational mode where output luminances are clipped at 100 nits. The bottom row shows the full dynamic range of the headset. The right-hand column in Figure 2 shows the view through the lens with a 5 stop digital reduction in exposure¹. Note how the specular highlights in the low dynamic range mode are clipped, whereas they are preserved in the HDR mode. This additional visual information (which is best observed in person) supports the wider range of luminance that we target with this demonstrator. Refer to our supplementary video to view these images in motion.

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¹These images were captured using a Red Komodo and Sigma EX DG fisheye lens at f/3.5, 320 iso, and 41 millisecond exposure time.